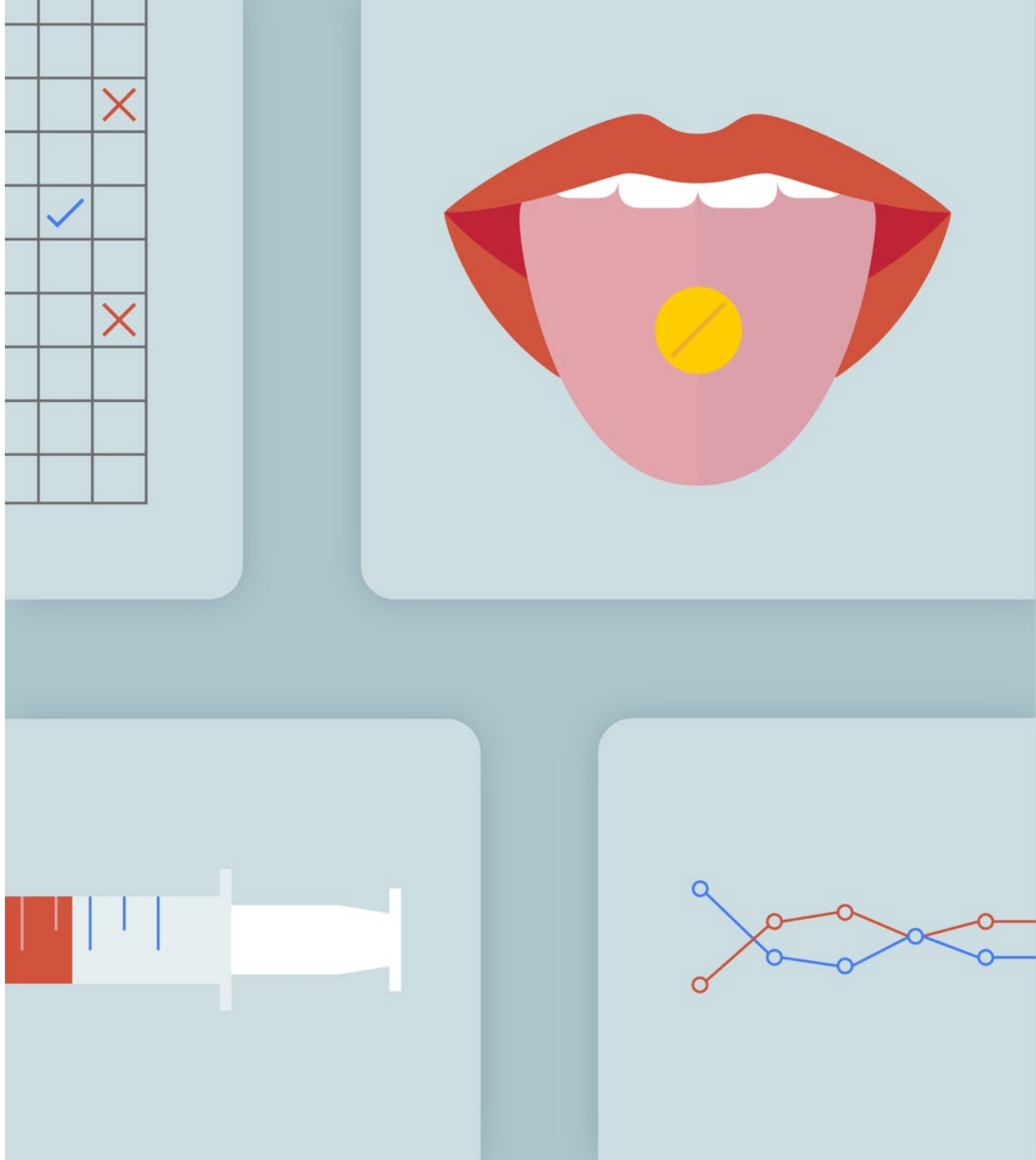
Dr. Sophie Bots (she/her)
Assistant Professor

R crash course

session 1



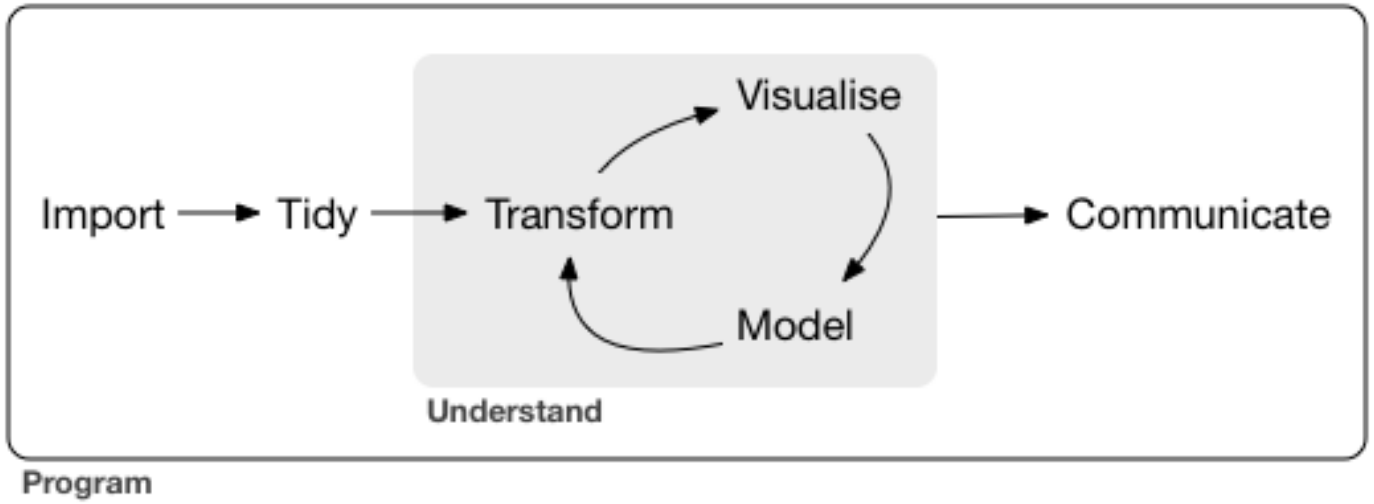
Utrecht University

Pharmacoepidemiology
and Clinical Pharmacology

Welcome!

Who are we?

What will we do during this course?



What can you do at the end of this course?

- Use RStudio for all tasks needed to complete the research project
- Know where and how to look for help (online, reference works) when you get stuck

<u>What this course will not cover</u>

- Underlying epidemiological and statistical theory
- More advanced programming techniques (*loops*, *if-else* statements, functions)

Learning goals

1. Perform basic coding tasks in RStudio like creating variables, and indexing data frames
2. Apply good workflow practices for coding, scripting, and getting help
3. Explain the rationale behind the tidy data framework and why this benefits data analysis
4. Transform datasets with the *dplyr* package
5. Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)
6. Perform descriptive and statistical analyses and interpret the results

What to expect

- Sessions with their own theme based on the learning goals

- Each session has preparation and revision to be completed outside of class; this builds your practical experience with RStudio so be sure to complete them and bring your questions to class

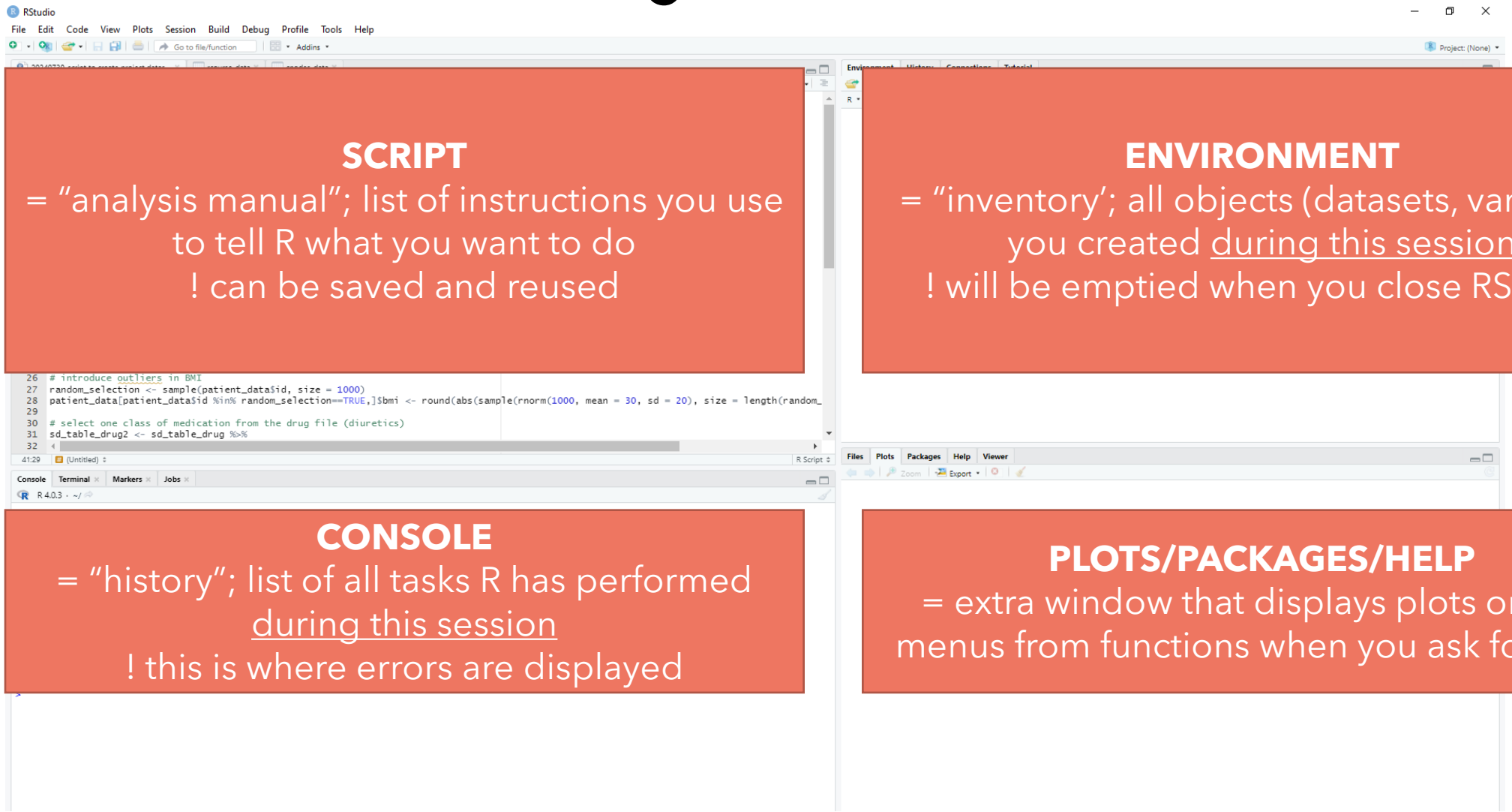
➔ check **course outline** for instructions

- Combination of mini-lecture (max. 30 mins) and practicals

Preparation for session 1

walkthrough

What are we looking at?



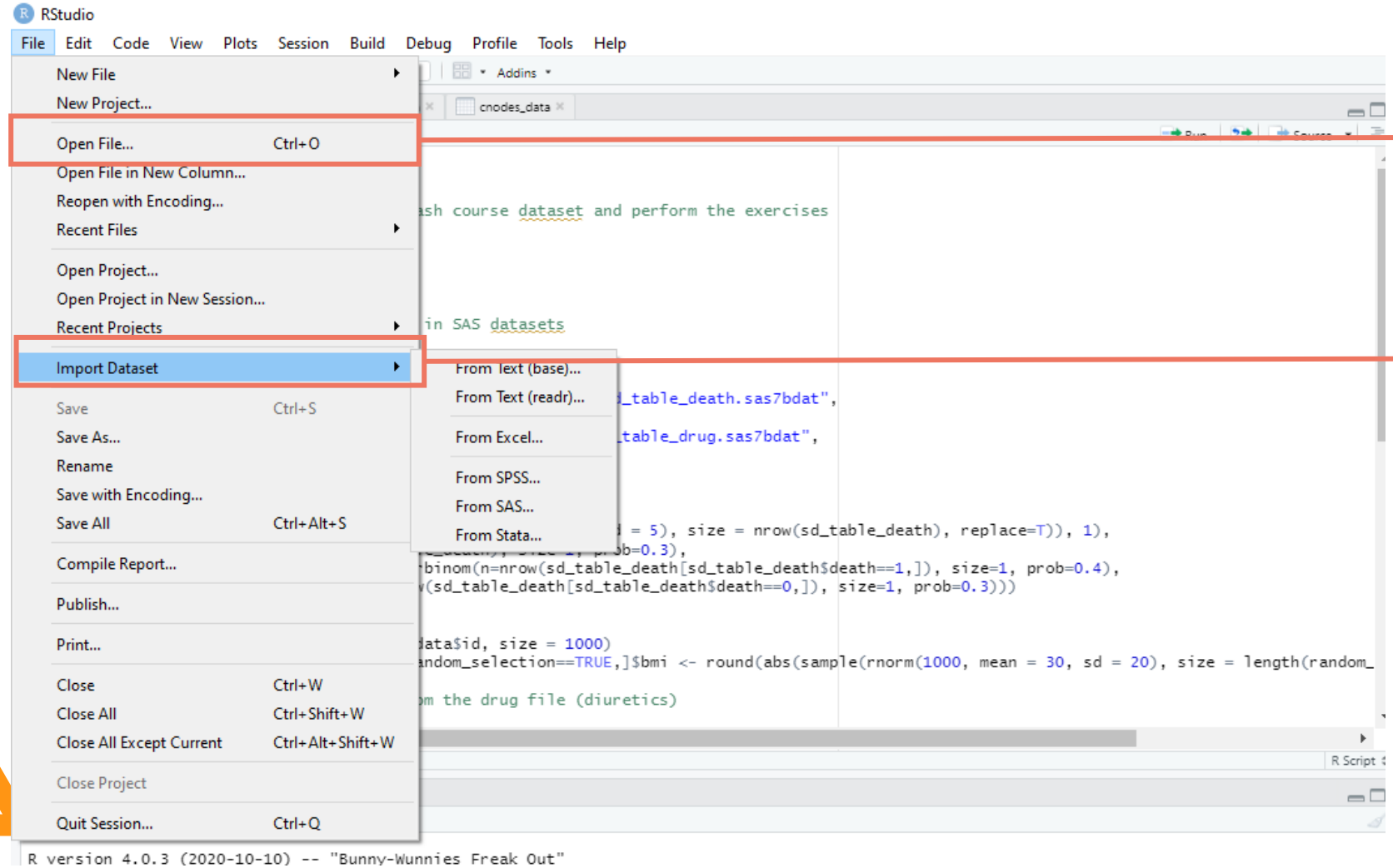
SCRIPT
= "analysis manual"; list of instructions you use to tell R what you want to do
! can be saved and reused

ENVIRONMENT
= "inventory'; all objects (datasets, variables) you created <u>during this session</u>
! will be emptied when you close RStudio

CONSOLE
= "history"; list of all tasks R has performed <u>during this session</u>
! this is where errors are displayed

PLOTS/PACKAGES/HELP
= extra window that displays plots or help menus from functions when you ask for them

1 – opening datasets in RStudio; option 1



for scripts

for R datasets (.Rdata)

for other data files (.csv, .xlsx, .txt, …)

Active break

Watch walk-though video if you haven't already

Open RStudio and return to wherever you stopped the preparation exercise for this session

With your current knowledge, try to complete at least one of exercises 6-8

Re-visit preparation materials if needed

1. ***Perform basic coding tasks in RStudio like creating variables and indexing data frames*** ***(preparation)***
2. Apply good workflow practices for coding, scripting, and getting help
3. Explain the rationale behind the tidy data framework and why this benefits data analysis
4. Transform datasets with the *dplyr* package
5. Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)
6. Perform descriptive and statistical analyses and interpret the results

Learning goals for today

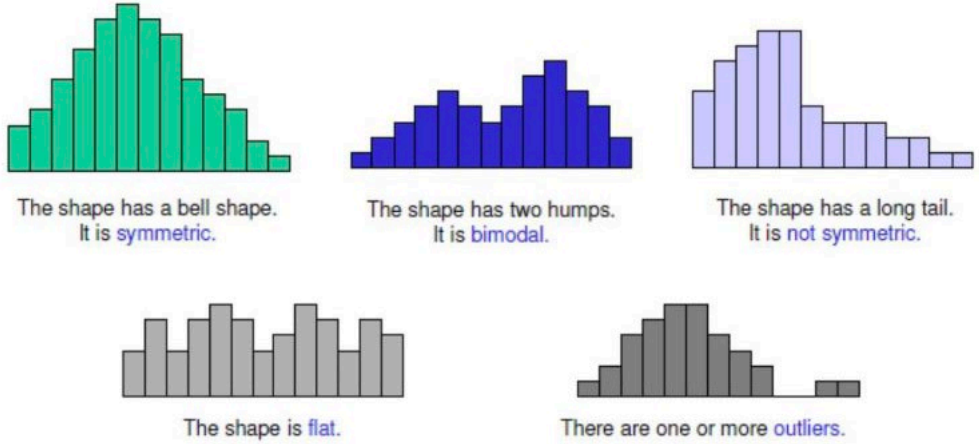
1. *Perform basic coding tasks in RStudio like creating variables, and indexing data frames (preparation)*
2. Apply good workflow practices for coding, scripting, and getting help
3. **Explain the rationale behind the tidy data framework and why this benefits data analysis**
4. Transform datasets with the *dplyr* package
5. **Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)**
6. Perform descriptive and statistical analyses and interpret the results

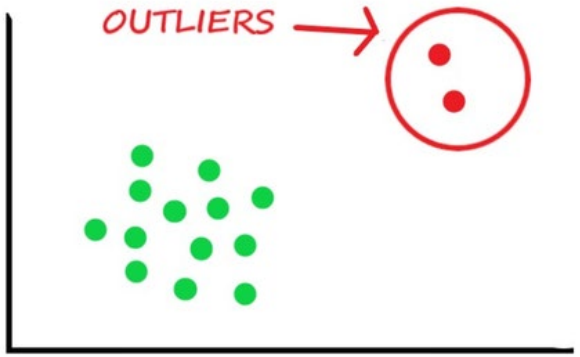
Perform exploratory data analysis with *ggplot2*

Part 1

learning goal 5

Why do exploratory analyses?





https://ouzhang.me/blog/outlier-series/outliers-part1/
https://leanscape.io/data-distributions-explained-what-are-the-different-types-of-distribution/

What kind of things to calculate?

Content:	presence/absence of variables measurement levels, completeness
Information:	numbers, averages (mean, median), distribution, spread, outliers
Correlation:	associations

What kind of things to calculate?

Content:	presence/absence of variables measurement levels, completeness
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Visual exploration

Distribution

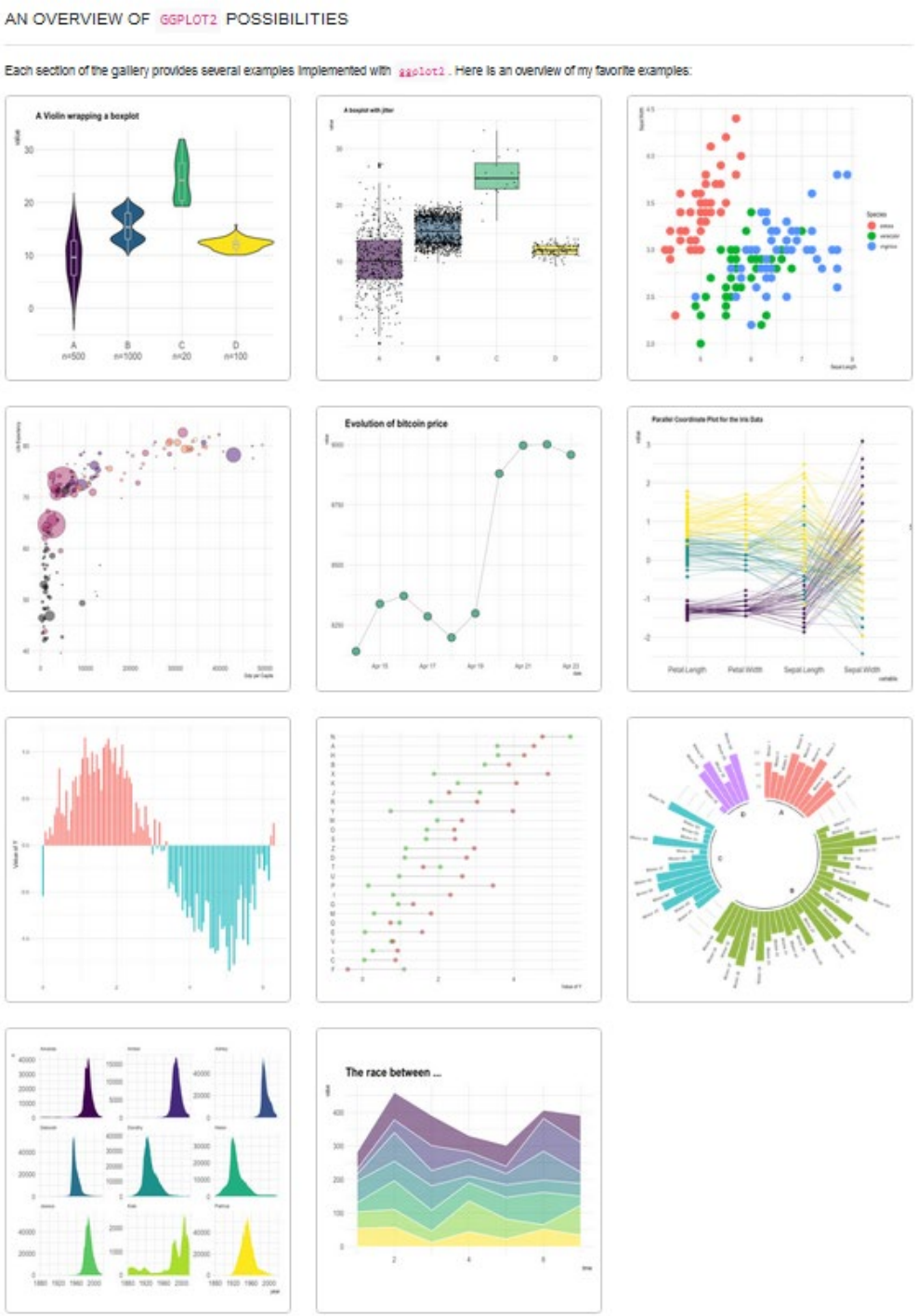
Violin | Density | Histogram | Boxplot | Ridgeline

Correlation

Scatter | Heatmap | Correlogram | Bubble | Connected scatter | Density 2d

Ranking

Barplot | Spider / Radar | Wordcloud | Parallel | Lollipop | Circular Barplot

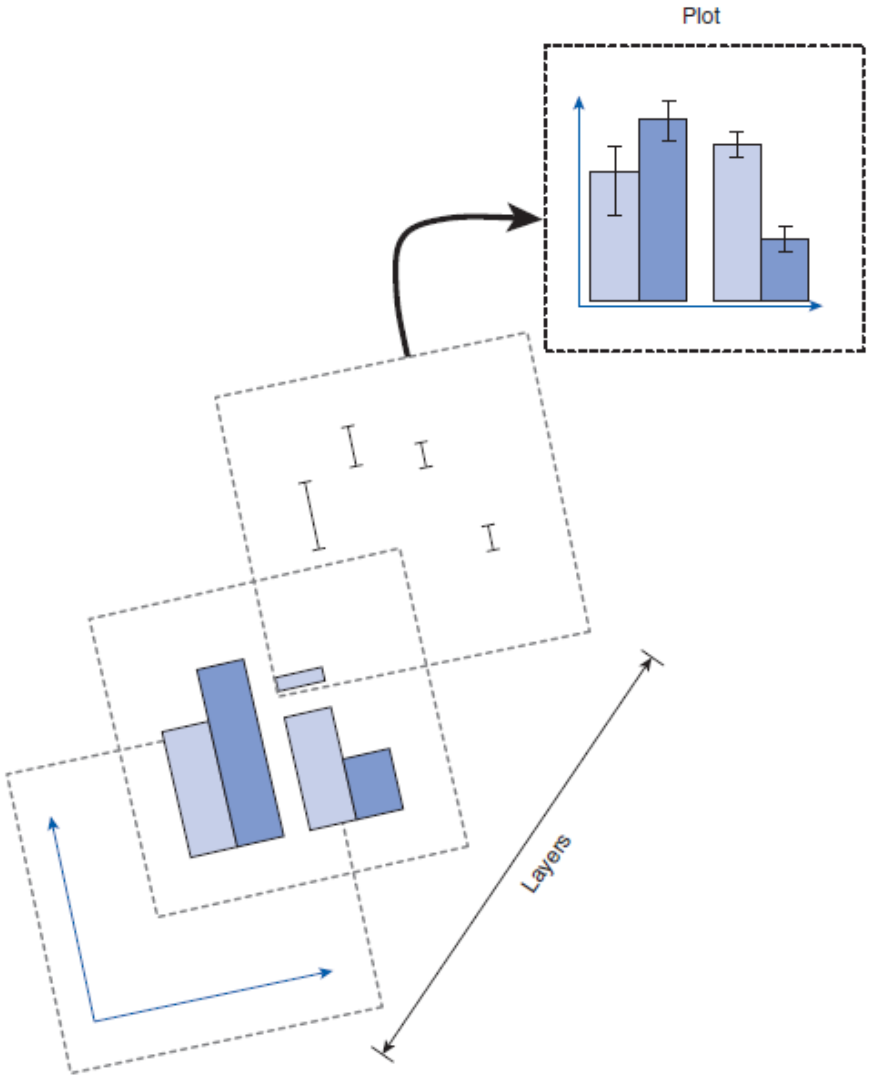


[The R Graph Gallery]

How does ggplot2 work?

ggplot2 creates plots based on layers

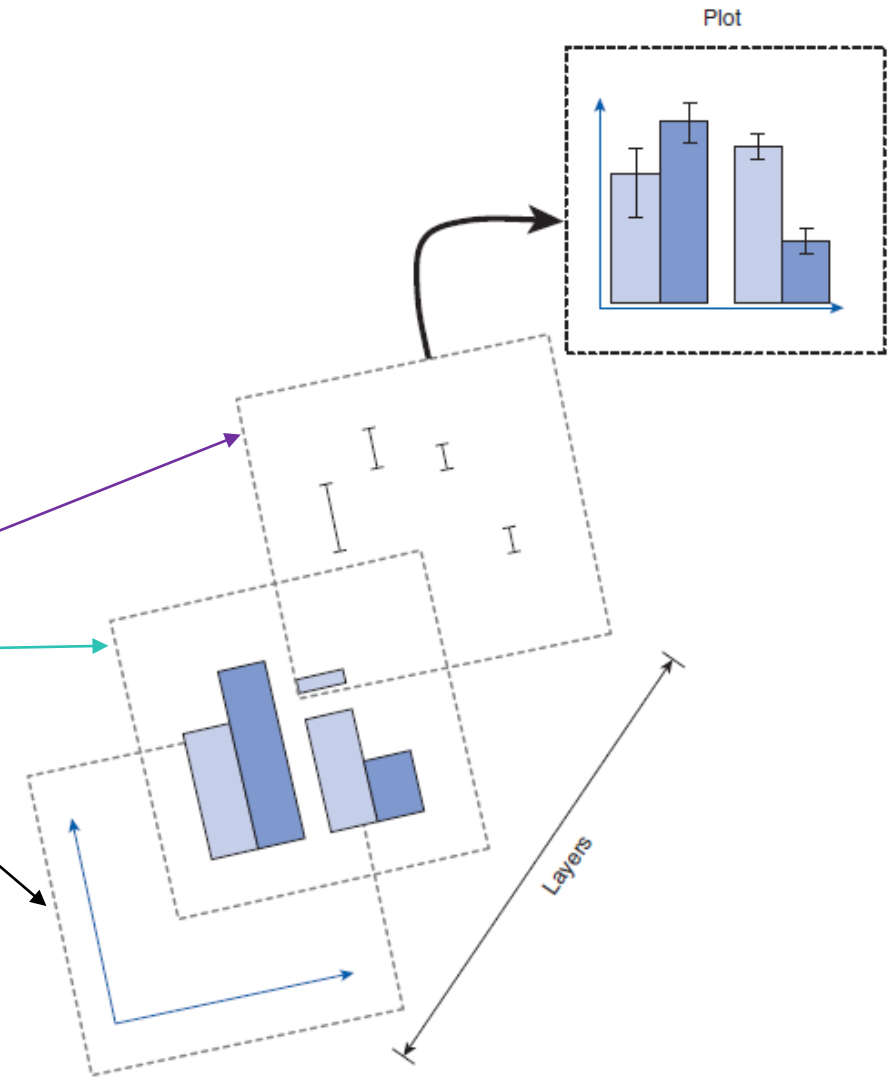
```
ggplot(data = <DATA>) +
        <GEOM_FUNCTION>() +
        <GEOM_FUNCTION>() +
        <…>
```



How does ggplot2 work?

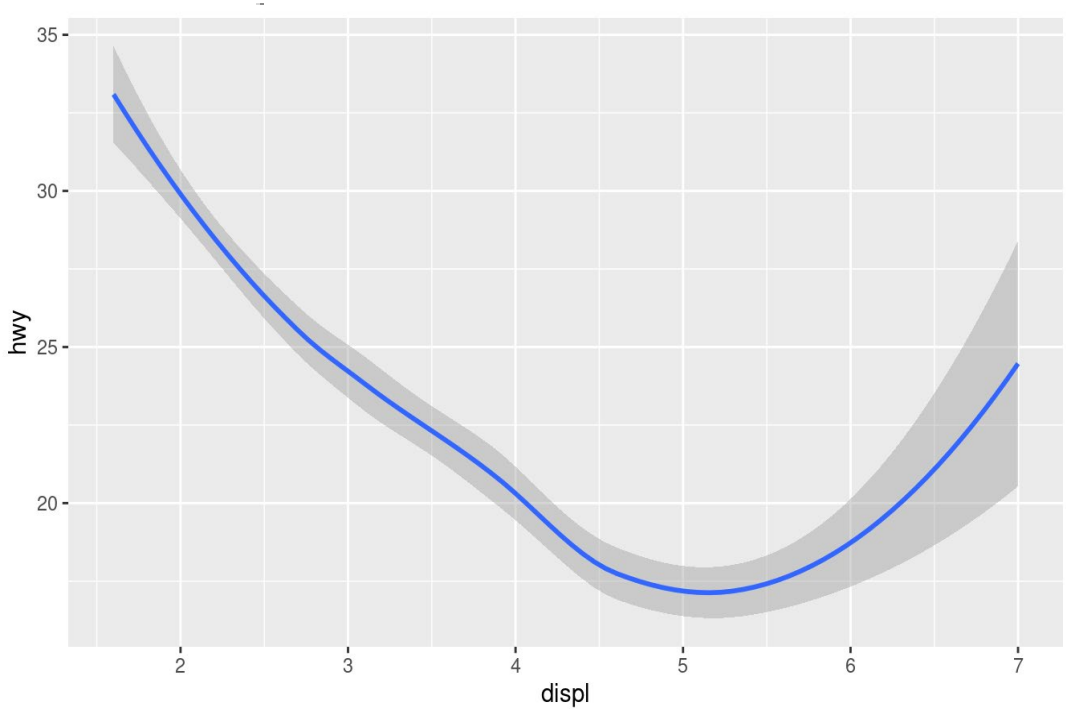
ggplot2 creates plots based on layers

```
ggplot(data = project_dataset) +
    geom_bar() +
    geom_errorbar()
```

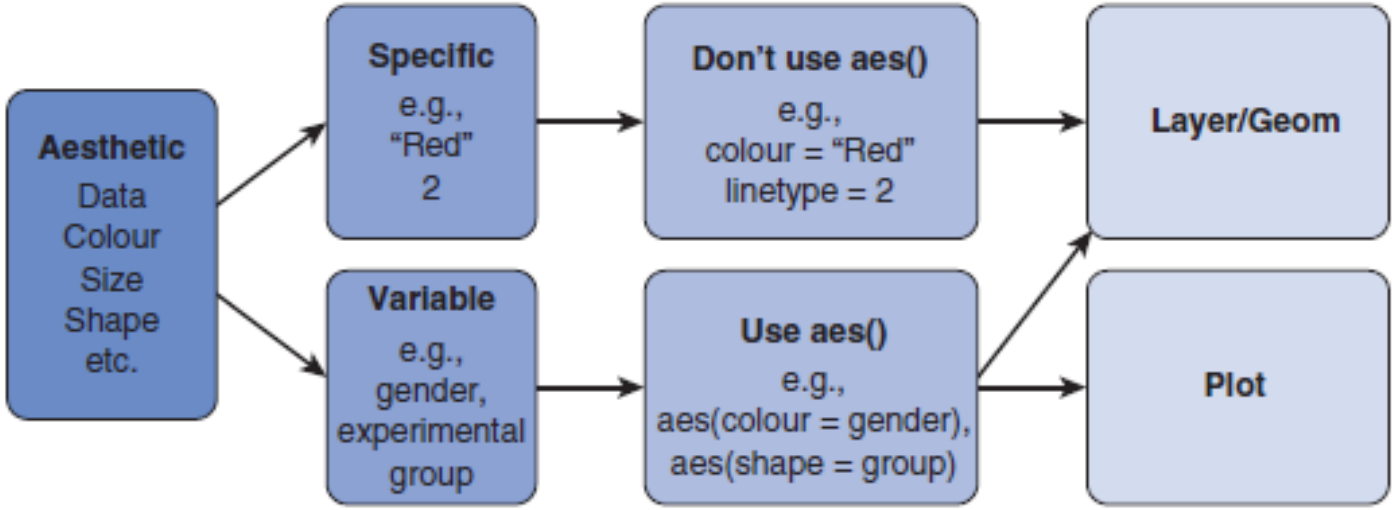


Pick your favourite type…

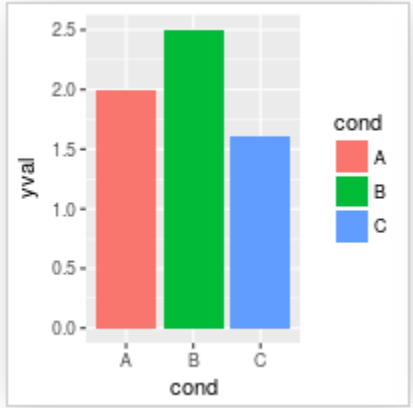
- scatterplot *geom_point*
- barplot *geom_bar*
- histogram *geom_histogram*
- line plot *geom_line*
- boxplot *geom_boxplot*
- smooth line *geom_smooth*



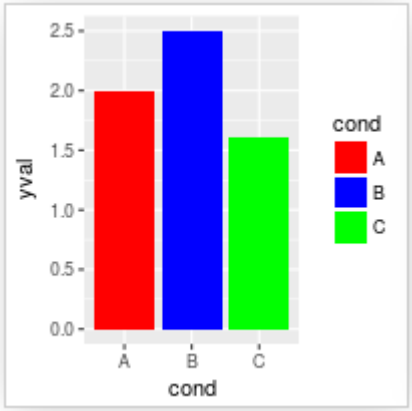
… and make it pretty



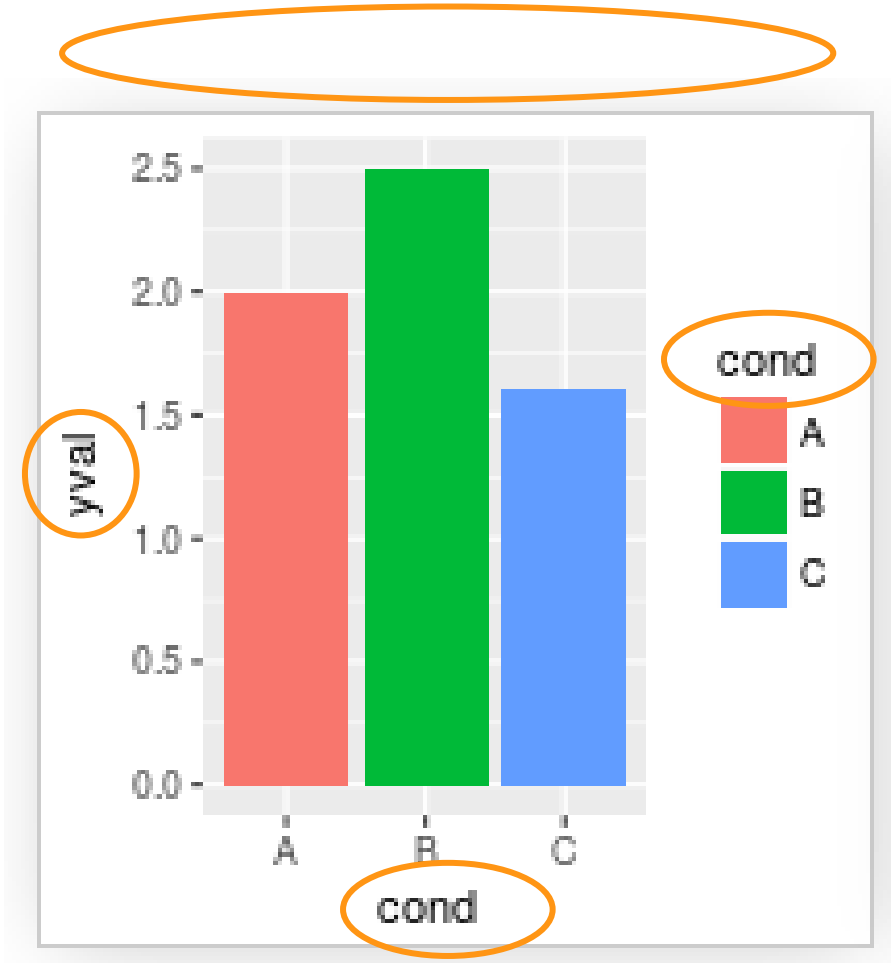

```r
# Bars: x and fill both depend on cond2
ggplot(df, aes(x=cond, y=yval, fill=cond)) + geom_bar(stat="identity")
```



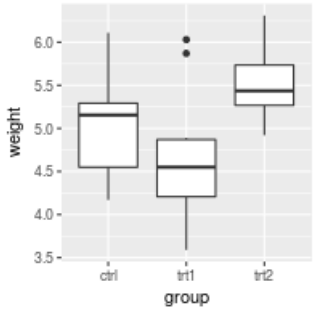

```r
ggplot(df, aes(x=cond, y=yval, fill=cond)) + geom_bar(stat="identity") +
    scale_fill_manual(values=c("red", "blue", "green"))
```

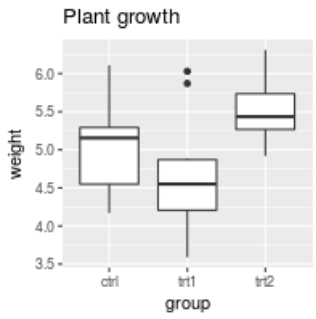


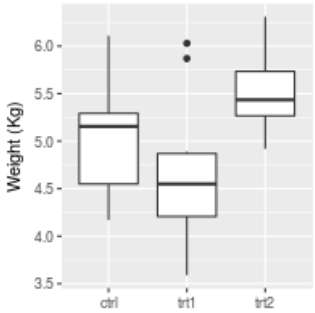
Can you explain what this figure shows?

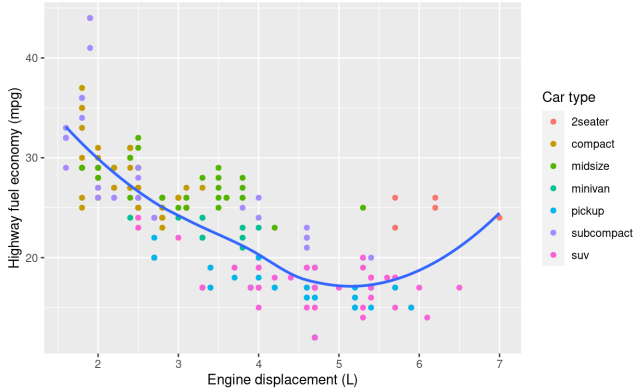


Figures should be self-explanatory









ggtitle() add title to your figure

```
bp + ggtitle("Plant growth")
```

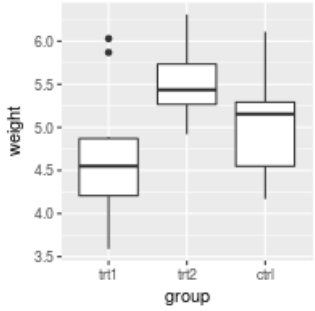
xlab() & *ylab()* add x- and y-axis labels

```
bp +  ylab("Weight (Kg)")
```

labs(…) combine labels in one statement

```
ggplot(mpg, aes(displ, hwy)) +
  geom_point(aes(colour = class)) +
  geom_smooth(se = FALSE) +
  labs(
    x = "Engine displacement (L)",
    y = "Highway fuel economy (mpg)",
    colour = "Car type"
  )
```

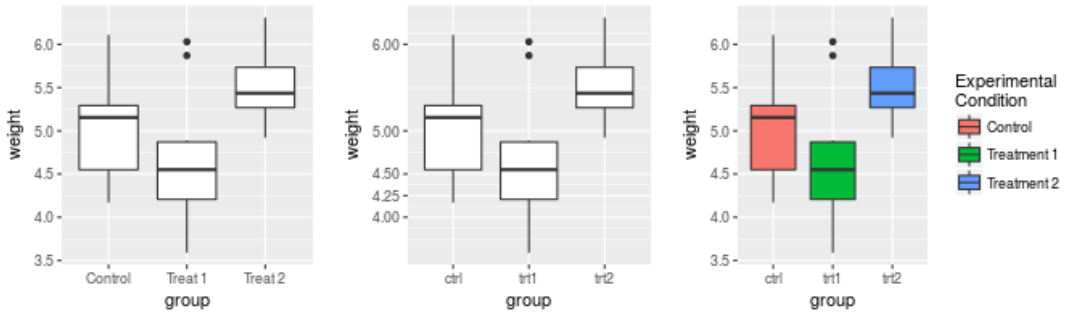
There is more …




```
bp + scale_x_discrete(breaks=c("ctrl", "trt1", "trt2"),
                      labels=c("Control", "Treat 1", "Treat 2"))
```

```
bp + scale_y_continuous(breaks=c(4, 4.25, 4.5, 5, 6,8))
```

```
bp + scale_fill_discrete(name="Experimental\nCondition",
                         breaks=c("ctrl", "trt1", "trt2"),
                         labels=c("Control", "Treatment 1", "Treatment 2"))
```



- *scale* options to change tick marks, colours, legend
- *xlim()*, *ylim()*, *coord_cartesian()* to set x- and y-axis limits
- *theme()* to change the overall look of your figure
- *guides()* to edit the legend

Active break

Go back to the graphs you created

Play around with them to see if you can:

- make a new type of graph
- change colours or other aesthetics
- add a title or other explanatory information

Reference material:

R for Data Science Chapter 9

Cookbook for R Chapter 8

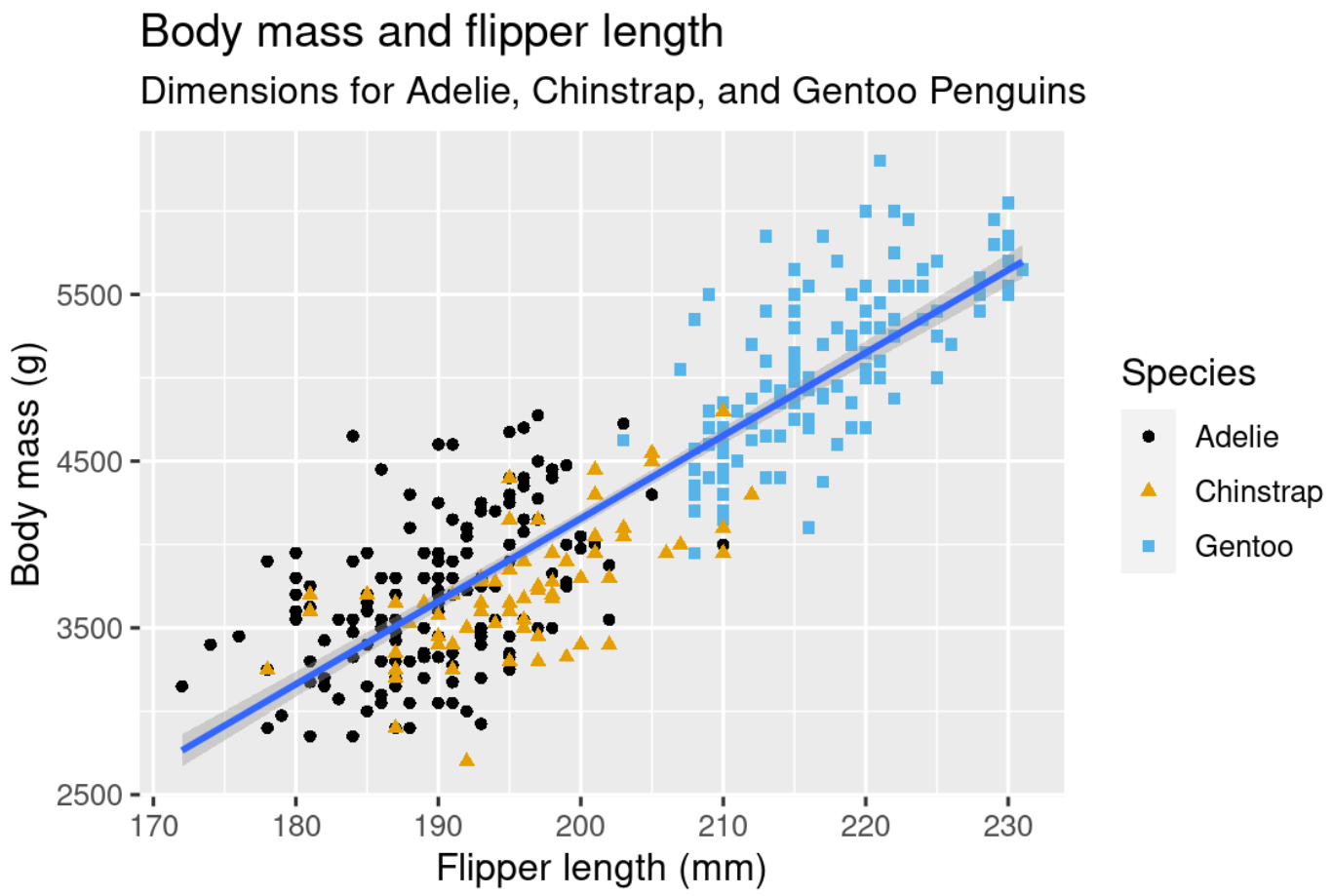
Ten Simple Rules for Better Figures

Interpret exploratory data analysis results

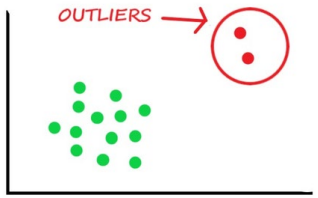
Part 1

learning goal 5

Okay… but what does it mean?



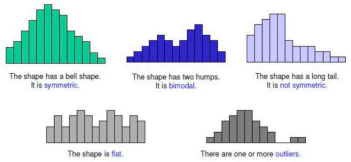
Things to think about

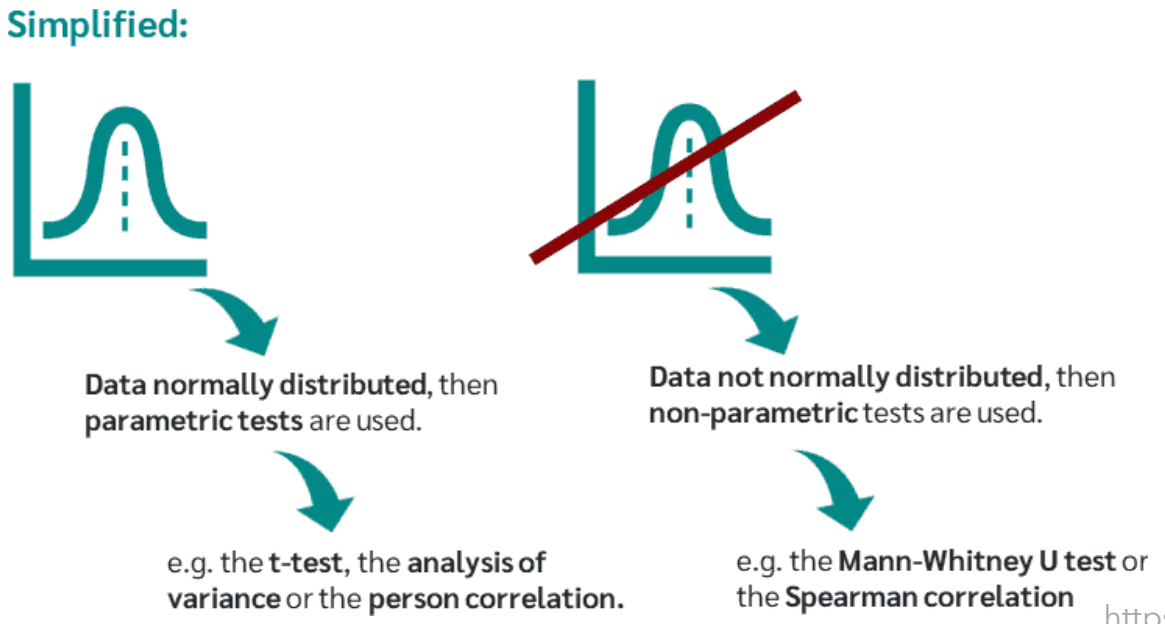


When is something an outlier?

Remove or correct?

How to deal with missing values?





https://datatab.net/tutorial/parametric-and-non-parametric-tests

Active break

Look at your exploratory figure(s)

What conclusions can you draw about your data?

The *tidy* data framework

Part 2

learning goal 3

Welcome to the tidyverse!

> "Tidy datasets are all alike, but every messy dataset is messy in its own way." — Hadley Wickham

what framework – organise and analyse your data in a constistent manner

how *tidyverse* package

why intuitive way of working for first-time R users

tidyverse contains large collection of tools, functions, and support

tidy data makes optimal use of how R works (vectorised)

https://www.tidyverse.org/

Data component - *tibble*

what modern version of *data.frame*

why enhanced printing (more informative display)

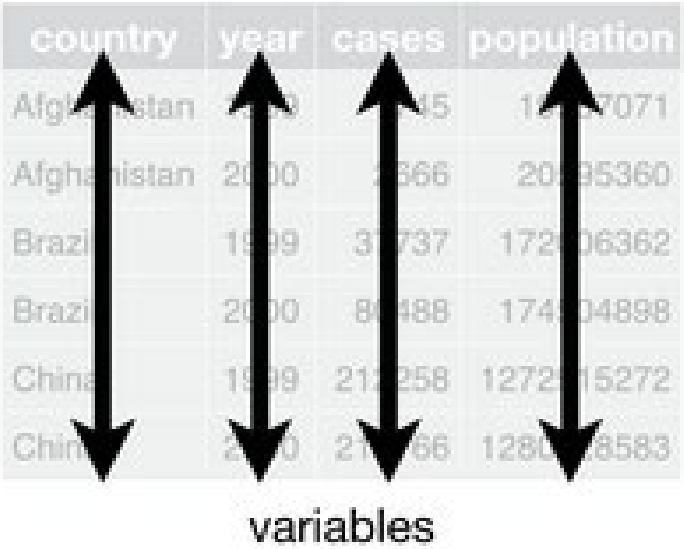
stricter than *data.frame*;

throws errors earlier

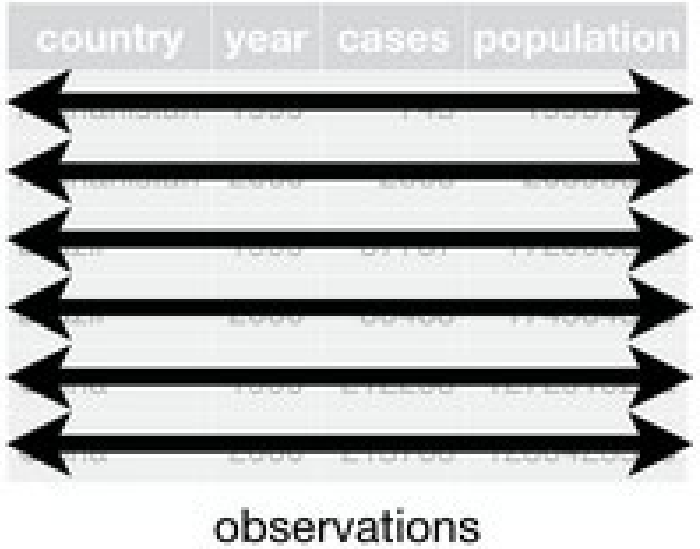
error/warning messages more detailed

What does ‘tidy’ data look like?

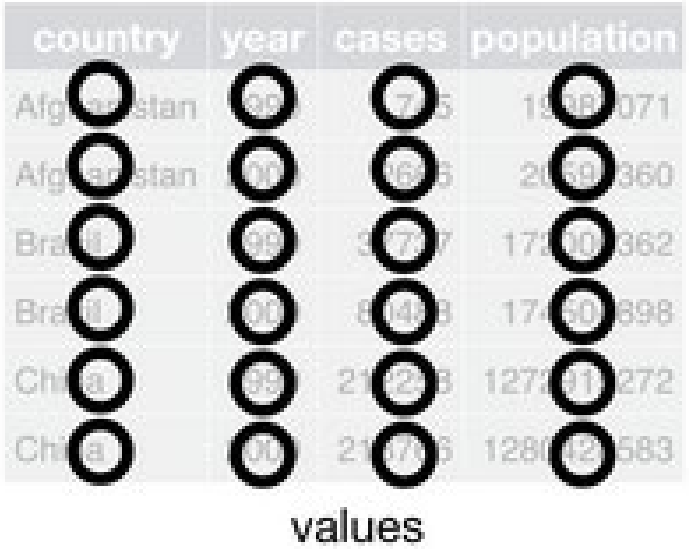
Each variable has own column



Each observation has own row



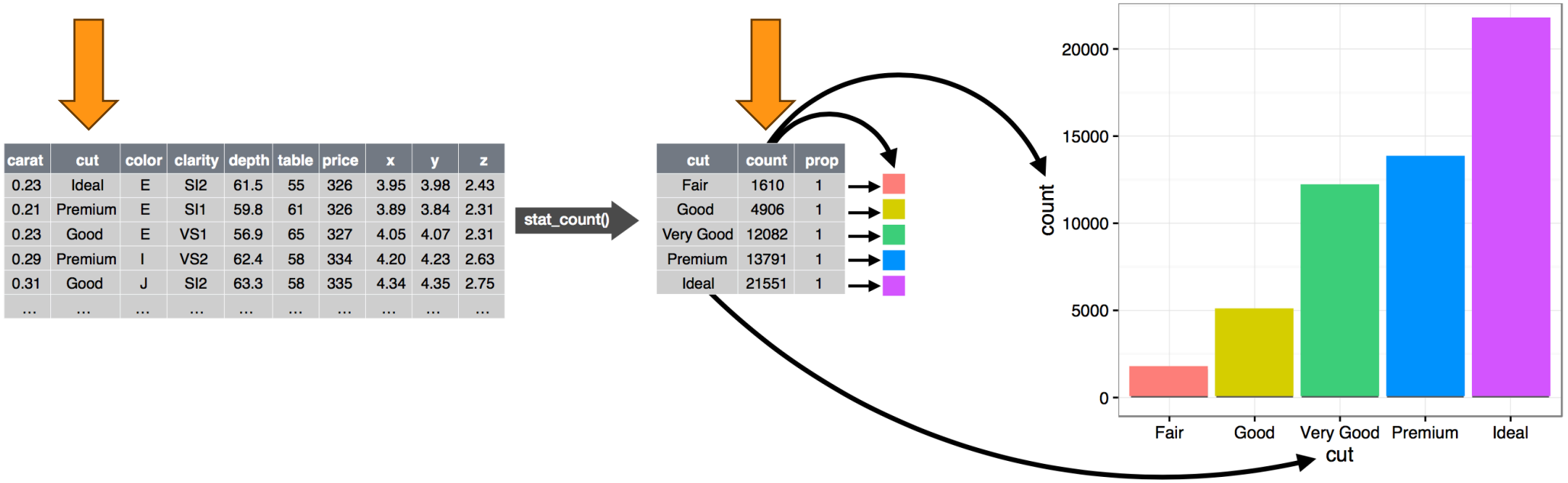
Each value has own cel



ggplot2 is also part of the tidyverse!

For example, *geom_bar()* counts the number of rows for a specific variable in the background to create a bar graph

This would not work if rows do not represent unique observations



Active break

Look at your dataset and discuss with you neighbour(s):

- is this dataset ‘tidy’?

- how can you check this?

- can you think of a dataset that is not ‘tidy’?

Reference material

R for Data Science Chapter 5

Learning objectives covered today

1. Perform basic coding tasks in RStudio like creating variables, and indexing data frames
2. Apply good workflow practices for coding, scripting, and getting help
3. **Explain the rationale behind the tidy data framework and why this benefits data analysis**
4. Transform datasets with the *dplyr* package
5. **Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)**
6. Perform descriptive and statistical analyses and interpret the results

Revise: check if you understood everything

R for Data Science

- Chapter 1 exercises
- Chapter 5 exercises section 5.2

Workflow chapters from *R for Data Science* (2, 4, 6, 8)

See course outline for additional revision materials and other optional further reading

Don't skip the *Workflow* part!

1. Perform basic coding tasks in RStudio like creating variables, and indexing data frames
2. **Apply good workflow practices for coding, scripting, and getting help (revision)**
3. Explain the rationale behind the tidy data framework and why this benefits data analysis
4. Transform datasets with the *dplyr* package
5. Visually explore datasets with *ggplot2* and interpret the results (exploratory analysis)
6. Perform descriptive and statistical analyses and interpret the results

See you next session!